

| Total   |
|---------|
| 17 / 18 |

1. a Opposite vectors:

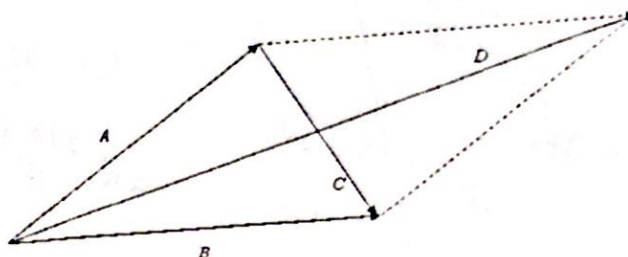
- ☒ a. have the same magnitude  
☐ b. have the same magnitude and direction  
☐ c. have the same direction  
☐ d. have different magnitudes and directions

2. b Adding a vector to the zero vector

- ☐ a. produces the zero vector  
☒ b. preserves the non-zero vector  
☐ c. changes the vector's direction  
☐ d. reduces the vector's magnitude

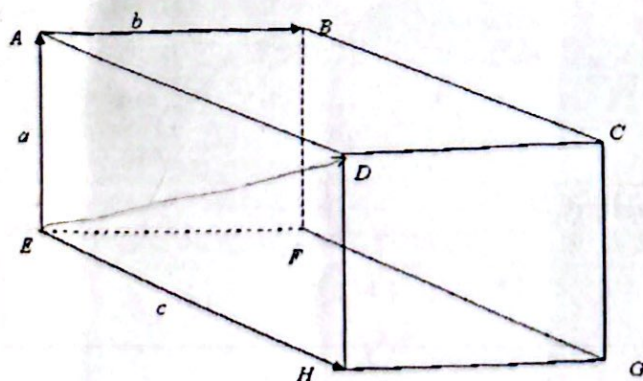
3. b Vector  $\vec{C}$  represents:

- ☐ a.  $\vec{A} + \vec{B}$   
☒ b.  $\vec{B} - \vec{A}$   
☐ c.  $\vec{C} + \vec{B}$   
☐ d.  $\vec{A} + \vec{C}$



4. d Express  $\vec{ED}$  by combinations of the vectors  $\vec{a}$ ,  $\vec{b}$  and  $\vec{c}$ .

- ☐ a.  $-\vec{b}$   
☐ b.  $\vec{a} - \vec{c}$   
☒ c.  $\vec{a} + \vec{b}$   
☒ d.  $\vec{a} + \vec{c}$



5. If  $\vec{x} = \vec{i} - \vec{j} + 2\vec{k}$ ,  $\vec{y} = 3\vec{i} + \vec{j} - 7\vec{k}$  and  $\vec{z} = 3\vec{j} + \vec{k}$ , simplify  $2(\vec{x} - \vec{y}) + \vec{z}$ ?

$$= 2[(\vec{i} - \vec{j} + 2\vec{k}) - (3\vec{i} + \vec{j} - 7\vec{k})] + (3\vec{j} + \vec{k})$$

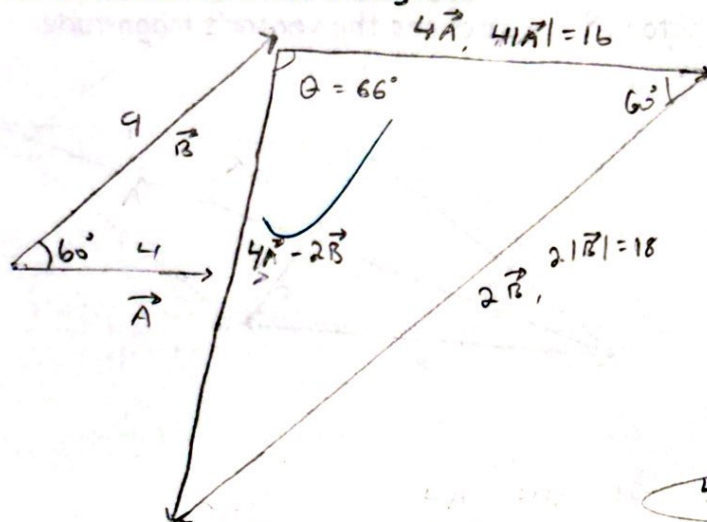
$$= 2(\vec{i} - \vec{j} + 2\vec{k} - 3\vec{i} - \vec{j} + 7\vec{k}) + 3\vec{j} + \vec{k}$$

$$= 2(-2\vec{i} - 2\vec{j} + 9\vec{k}) + 3\vec{j} + \vec{k}$$

$$= -4\vec{i} - 4\vec{j} + 18\vec{k} + 3\vec{j} + \vec{k}$$

$$= -4\vec{i} - \vec{j} + 19\vec{k} = 19\vec{k} - 4\vec{i} - \vec{j}$$

6. a)  $|\vec{A}| = 4$ ,  $|\vec{B}| = 9$ , and the angle between  $\vec{A}$  and  $\vec{B}$  is  $60^\circ$ . Determine  $|4\vec{A} - 2\vec{B}|$  and its direction. Provide a diagram.



$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$|4\vec{A} - 2\vec{B}| = \sqrt{16^2 + 18^2 - 2(16)(18)\cos 60}$$

$$= \sqrt{292}$$

$$= 17.1$$

$$\frac{\sin \theta}{18} = \frac{\sin 60}{17.1}$$

$$\theta = \sin^{-1}\left(\frac{18 \sin 60}{17.1}\right)$$

$$\theta = 66^\circ$$

$$4\vec{A} - 2\vec{B} = 17.1 [66^\circ \text{ from } \vec{A}]$$

b) Determine a unit vector in the same direction as  $4\vec{a} - 2\vec{b}$

$$\frac{1}{|4\vec{a} - 2\vec{b}|} \cdot (4\vec{a} - 2\vec{b})$$

$$= \frac{1}{17.1}$$

7. Express the sum as single vector.

$$\vec{AB} - \vec{CB} - \vec{AD} + \vec{CD}$$

$$= \vec{AB} + \vec{BC} + \vec{DA} + \vec{CD}$$

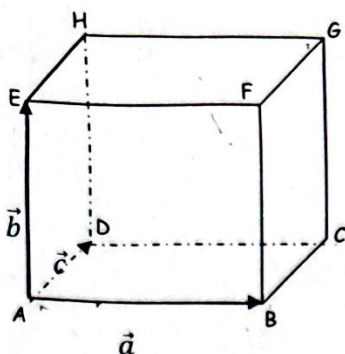
$$= \vec{AB} + \vec{BC} + \vec{CD} + \vec{DA}$$

$$= \vec{AA}$$

$$= \vec{0}$$



8. A cube is constructed from the three vectors,  $\vec{a}$ ,  $\vec{b}$ , and  $\vec{c}$  as shown below.



Express each diagonal in terms of  $\vec{a}$ ,  $\vec{b}$ , and  $\vec{c}$

a)  $\vec{AG}$   
 $= \vec{a} + \vec{b} + \vec{c}$

b)  $\vec{BH}$   
 $= \vec{b} + \vec{c} - \vec{a}$

9. Using the cube above, write the following as a single vector:

$$\vec{CB} - \vec{GC} + \vec{BA} - \vec{GF}$$

$$= \vec{CB} + \vec{CG} + \vec{BA} + \vec{FG}$$

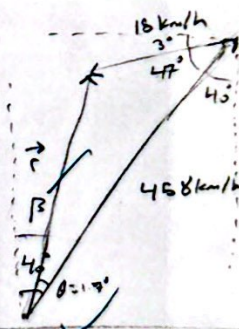
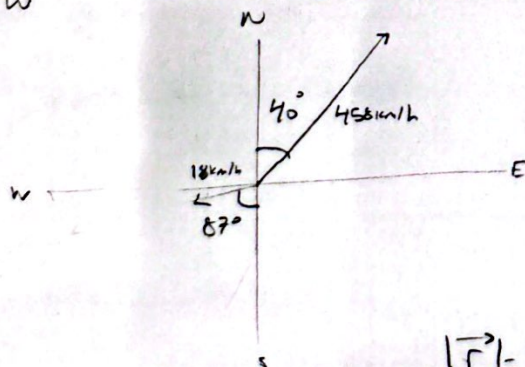
$$= \vec{CB} + \vec{BF} + \vec{FE} + \vec{EH}$$

$$= \vec{CH}$$

10. An airplane is flying N40°E at a speed of 458 km/h when it encounters a wind S87°W of 18 km/h. Determine the ground velocity of the airplane.

Include a diagram in your solution.

Wind going towards direction S87°W



$$|\vec{r}| = \sqrt{458^2 + 18^2 - 2(458)(18)\cos 47}$$

$$|\vec{r}| = 445.92 \text{ km/h}$$

$$\frac{\sin \theta}{18} = \frac{\sin 47}{445.92}$$

$$\sin \theta = \frac{18 \sin 47}{445.92}$$

$$\theta = \sin^{-1} \left( \frac{18 \sin 47}{445.92} \right)$$

$$\theta = 1.7^\circ$$

$$\beta = 40^\circ - 1.7^\circ = 38.3^\circ = 38^\circ$$

$$\vec{r} = 445.92 \text{ km/h [N } 38^\circ \text{ E]}$$